



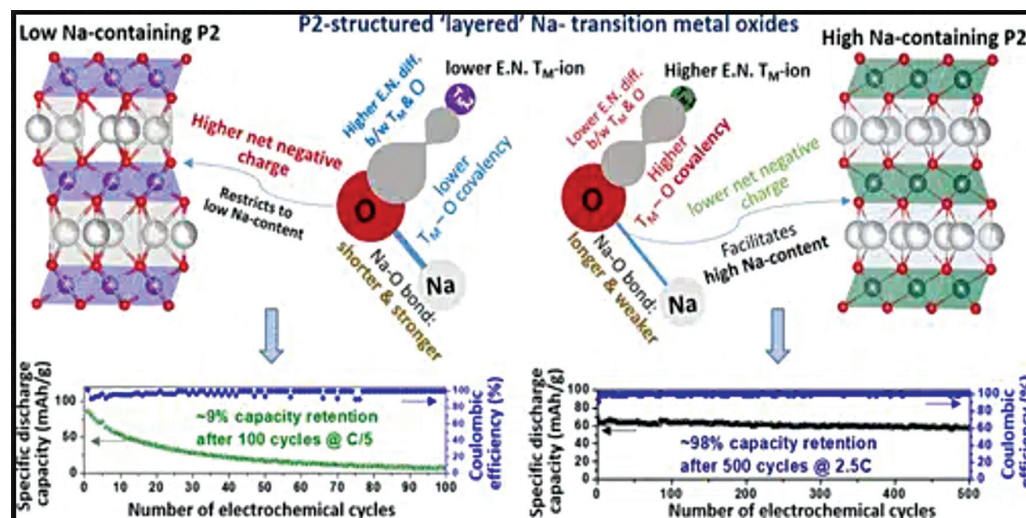
RESEARCH & INNOVATION UPDATES

Functional memory designed with tin oxide for high-efficiency computers

Researchers at the Centre for Nano and Soft Matter Sciences (CeNS) have designed a functional memory using tin oxide slanted nanorod arrays with high potential for high-density and high-efficiency computing systems. Tin oxide nanostructures were prepared using the glancing angle deposition technique with electron-beam evaporation. Memory devices showed low voltage, average on/off ratio, long endurance, and better retention with self-compliance

effect in the dark. An unusual negative photo response was observed with an enlarged on/off ratio greater than 107 and a faster response time under different illuminations. The devices undergo electrical set and optical reset, with voltage bias for the low resistance state and light exposure for the high resistance state. Multiple resistance states are achieved via current and optical modulation, driven by oxygen vacancies for optical resistance switching.

High-performance cathode materials for Ni-ion batteries developed at IIT Bombay



Depiction of the influence of raised T_M -O bond covalency on the stabilisation of the P2 structure for $Na-T_M$ -oxide even at a high Na-content of ~ 0.84 p.f.u.; thus, leading to the development of a high capacity (greater than most reported to-date) and very stable P2-structured $Na([Li_{0.06}Li_{0.04}Mg_{0.02}Ni_{0.22}Mn_{0.65}]O)_x$ cathode material, which exhibits excellent performances in Na-ion cells (as published in Chem. Mater. 34[23] (2022) 10470–10483)

Researchers at the Indian Institute of Technology (IIT) Bombay have developed new, high-performance cathode materials for Na-ion batteries that address air/water and structural-electrochemical instability issues at the same time. The team used materials science and electrochemical principles to develop a universal criterion for high-perfor-

mance, stable Na-ion battery cathodes. In the $Na-T_M$ -oxide structure, alternating NaO_2 and TMO_2 'slabs' share negatively charged O-ions, ionic for Na-O and ionic-covalent for TM-O bonds. By tuning TM-O bond covalency through TM-layer cation combination, electrostatic attraction and repulsion between Na- and O-ions across the Na-layer can be adjusted. Increasing O-ion's negative charge enhances Na-O bond strength,

stabilising $Na-T_M$ -oxide cathodes. Water-stable cathodes for Na-ion batteries can be made using a cost-effective aqueous process. Stronger TM-O covalency weakens Na-O bonds and widens inter-slab spacing. Lowering O-ion's negative charge improves rate-capability and power density in Na-ion battery cathodes.